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Performance improvement of photovoltaic power systems using an optimal control strategy based on whale optimization algorithm

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ABSTRACT

Photovoltaic (PV) installations are consistently increasing all over the world, leading to a high penetration to the electric grid. Tremendous efforts should be exerted to maintain the operation of the PV systems at optimal conditions. This paper introduces an optimal control strategy with the purpose of enhancing the performance of PV systems. This control strategy is based on the proportional-integral (PI) controller, which is designed by using the whale optimization algorithm (WOA). The response surface methodology (RSM) model is established to create the objective function and its constraints. The proposed WOA-based PI controllers are utilized to control the DC chopper and grid-side inverter in order to achieve a maximum power point tracking operation and improve the dynamic voltage response of the PV system, respectively. The effectiveness of the control strategy is tested under different operating conditions of the PV system such as (1) subject the system to symmetrical and unsymmetrical fault conditions, (2) study the system responses under different irradiation and temperature conditions using real data extracted from a field test, and (3) subject the system to a sudden load disturbance in an autonomous operation. This effectiveness is compared with that achieved using the generalized reduced gradient (GRG) algorithm-based PI controller. The validity of the proposed control strategy is extensively verified by the simulation results, which are performed using PSCAD/EMTDC environment.

1. Introduction

Photovoltaic (PV) technology received a great interest worldwide over the last decades. It is expected that the solar PV will be the cheapest renewable power technology in the near future because of the deep cost reduction of the PV components [1]. This reflects the exerted significant efforts to enhance efficiency and develop the manufacturing process of the PV. The statistics of the global PV market state that the total PV installation in Europe can reach up to 156 GW in 2018, which represents a new record of PV installation [2]. This is an indicator to the high penetration of the PV systems into the electric grids. The integration of large scale PV systems with the electric grids can cause many dynamic and operation problems, which should be solved optimally to maintain the power system stability and reliability [3–7].

There are a lot of methods used to investigate and enhance the performance of the PV systems under different operating conditions. In Ref. [8], the reactive power and the terminal voltage of a grid-connected PV system are controlled during a grid disturbance using a neuro-fuzzy control strategy. Although, the neuro-fuzzy controller is efficiently capable of dealing with the system nonlinearity and parameters sensitivity, it depends mainly on the designer experience for selecting the artificial neural network (ANN) design and the fuzzy rules. Moreover, the training of the ANN takes a long time, which affects the dynamic performance. In Refs. [9,10] the low voltage ride through (LVRT) capability of PV systems is proposed using the proportional-integral (PI) controller. The voltage stability of PV systems is improved using a cascaded PI control scheme [11]. In Ref. [12], adaptive or self-tuning PI controllers are revealed to improve the dynamic performance of the PV power plants. However, the fixed-gains PI controllers are still the most commonly used in industry. Several studies have used this type of control to improve the dynamic performance of PV systems, but its design relies on the trial and error method, which is not an accurate method for the controller design [13–15]. Although, the PI controller offers a wide stability margin in modern control systems, it suffers from the system nonlinearity and the system's parameters variation. Therefore, the optimal design of such controllers in heavy nonlinear systems like PV systems is considered a cumbersome problem for the designers of power system control. The principal problem is the difficulty to represent the system model by linear transfer functions. Therefore, several optimization

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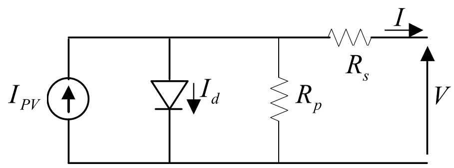

Fig. 1. The equivalent circuit of the single-diode PV model.

methods are used to solve such design problems such as genetic algorithm (GA) [16], the Taguchi approach [17], harmony search algorithm [18], and shuffled frog leaping algorithm [19]. The significant development of the computational evolutionary algorithms and its powerful application to solve complex nonlinear optimization problems represent the main motivation of the author to apply the novel whale optimization algorithm (WOA). It is a novel meta-heuristic optimization algorithm, which is motivated by the social behavior of the humpback whales. This algorithm easily describes the hunting process of the humpback whales to the prey. It was presented by Mirjalili and Lewis in 2016 [20]. The WOA possesses some merits over other swarm optimization algorithms such as its high convergence speed and its lower number of parameters to be designed. Recently, it was successfully applied to many classic mathematical optimization problems and various engineering problems with the purpose of design optimization [20]. To the best of the Author's knowledge, the proposed algorithm has not been reported so far in renewable energy systems literatures.

In this paper, a novel application of the WOA-based PI control strategy is proposed to enhance the dynamic performance of PV systems. The DC boost chopper of the PV system is implemented to extract the maximum power. The grid-side inverter of the PV system is used to control both of the DC-link voltage and the voltage at the point of common coupling (PCC). All power electronic devices are controlled using the proposed control strategy. To perform the optimization process, the objective function and its constraints are established using the response surface methodology (RSM), which is a powerful statistical method that expresses the system response in terms of the design variables through empirical formulas. In this study, the RSM model is a second order to obtain accurate results. The proposed WOA is applied to the RSM model as an offline optimization problem using MATLAB environment [21]. The validity of the proposed control strategy is extensively verified by the simulation results, which are performed using PSCAD/EMTDC environment [22]. The effectiveness of the proposed controller is checked under different operating conditions of the PV system such as (1) subject the system to symmetrical and unsymmetrical fault conditions, (2) study the system responses under different irradiation and temperature conditions using real data extracted from a field test, and (3) subject the system to a sudden load disturbance in an autonomous operation. This effectiveness is compared with that achieved using the GRG-based PI controller. With the WOA-based PI control strategy, the dynamic performance of PV systems should be enhanced.

The paper is structured as follows: Section 2 presents the PV system modeling. In Section 3, the converters control is introduced. Section 4 describes the problem formulation and the WOA. In Section 5, the design procedure is carried out. Section 6 presents the simulation results and discussion. Finally, Section 7 draws the conclusion.

2. PV system modelling

Fig. 1 indicates the equivalent circuit of the single-diode PV model, which is selected to model the PV module due to its accuracy and simplicity. The nonlinear current–voltage characteristic of the PV module can be expressed by the following expression [23,24]:

$$I = I_{PV} - I_o\left[\exp\left(\frac{V + R_s I}{aV_t}\right) - 1\right] - \frac{V + R_s I}{R_p} \qquad (1)$$

where I_{PV} is the photovoltaic current, I_o, a are the reverse bias current and ideality factor of the diode, respectively, R_s, R_p are the series and parallel resistances, and V_t is the thermal voltage.

I_{PV} can be written by the following equation:

$$I_{PV} = (I_{PV,n} + K_I \Delta T)\frac{G}{G_n} \qquad (2)$$

where $I_{PV,n}$ is the photovoltaic current under the nominal condition, K_I is the short circuit current per temperature factor, ΔT is the temperature difference between the actual and nominal values, G and G_n are the solar irradiation and its nominal value, respectively [23]. The mathematical model of $I_{PV,n}$ and I_o are written in details in Ref. [12]. A Kyocera KC200GT PV module is chosen for this study and its characteristics are demonstrated in Table 1 [25]. All the data of the PV model are depicted in Ref. [23]. For a large scale PV system, series-parallel connections of the PV modules are required [15]. In this study, a 2.5 MW PV system is used and its electrical characteristics are illustrated in Fig. 2a. The data of the PV system are listed in Table 2 [12]. The PV system is connected to the electric grid through a DC boost chopper, a DC-link capacitor of 10 mF, a grid-side inverter, step up transformers, and transmission lines, as shown in Fig. 2b.

Table 1
Characteristics of KC200GT PV module [23].

Short circuit current (I_{sc})	8.21 A
Open circuit voltage (V_{oc})	32.9 V
Current at maximum power point (I_{mp})	7.61 A
Voltage at maximum power point (V_{mp})	26.3 V
Maximum output power (P_{max})	200 W
K_V	−0.123 V/°C
K_I	0.00318 A/°C
Number of series connected cells (N_s)	54

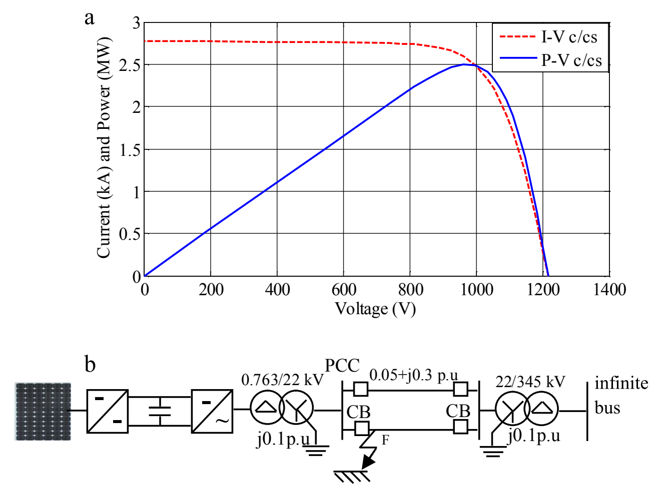

Fig. 2. (a) Electrical characteristics of a 2.5 MW PV system. (b) Grid-connected PV system.

Table 2
Data of a 2.5 MW PV system [12].

P_{max}	2.5 MW
V_{mp}	973.1 V
I_{mp}	2572.18 A

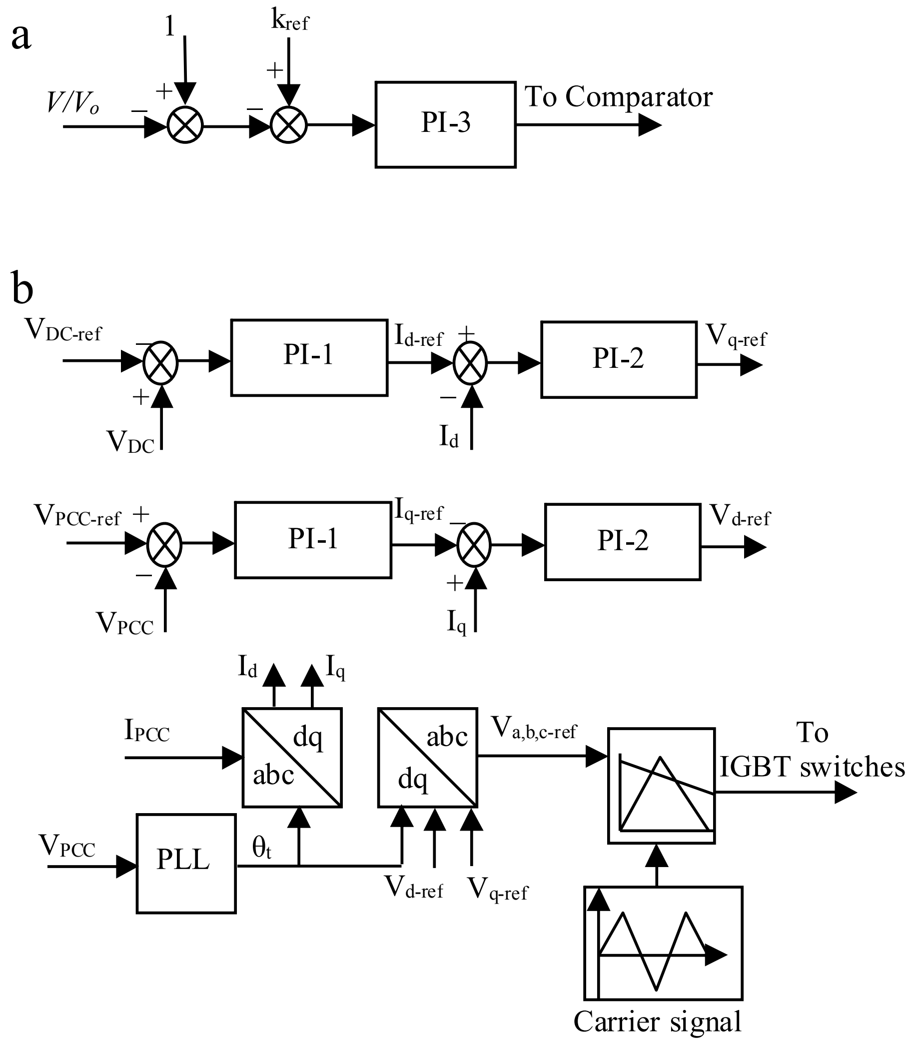

Fig. 3. (a) DC chopper control. (b) Grid-side inverter control.

3. Converters control

3.1. DC chopper

In this study, a DC boost chopper is utilized to control the PV voltage for tracking the maximum power point. The control scheme relies on controlling the duty cycle of the electronic switch of the chopper using the proposed WOA-based PI controller, as shown in Fig. 3a. The desired signal of the duty cycle is calculated as follows [12], [26–28]:

$$k_{ref} = 1 - \frac{N_M K_M V_{oc-pilot}}{V_o} \tag{3}$$

where N_M is the number of series connected modules in the PV system, K_M is a constant, $V_{oc-pilot}$ represents the open circuit voltage of the pilot module, V_o is the output voltage of the DC chopper circuit, and V is the output voltage of the PV system. The output signal of the proposed controller is then compared with a 4 kHz carrier signal of triangular waveform to produce the gate pulses of the insulated gate bipolar transistor (IGBT) switch.

3.2. Grid-side inverter

A grid-side inverter of a two level and three phase configuration is used in this study. It is implemented to control both of the DC-link voltage (V_{DC}) and the voltage at the point of common coupling (V_{PCC}) via a cascaded control scheme using the proposed WOA-based PI controller, as indicated in Fig. 3b. The three-phase voltages at the PCC are sensed and a phase locked loop (PLL) circuit is used to detect the angle of the frames transformation (θ_t). The reference signals $V_{a,b,c-ref}$ are compared with a 1 kHz carrier signal of triangular waveform to produce the gate pulses of the IGBT switches.

4. Problem formulation and the WOA

4.1. The RSM

The RSM is a powerful statistical method, which can be used to establish an empirical model by obtaining the relationship between the system response and its design variables [29,16]. The design of the PI controllers should be done under the system's worst condition. Therefore, a three-line to-ground (3LG) fault is applied to the PV system at point F, shown in Fig. 2b, as a network disturbance. Multiple simulation runs are performed using PSCAD/EMTDC to provide the system responses with variation of design variables. The transient response specifications such as the maximum percentage overshoot (MPOS) and the settling time (T_s) of the V_{PCC} are selected as the system responses. These specifications are changed by the design variables variant (PI controllers parameters). The RSM model exhibits a second order model and depends on the central

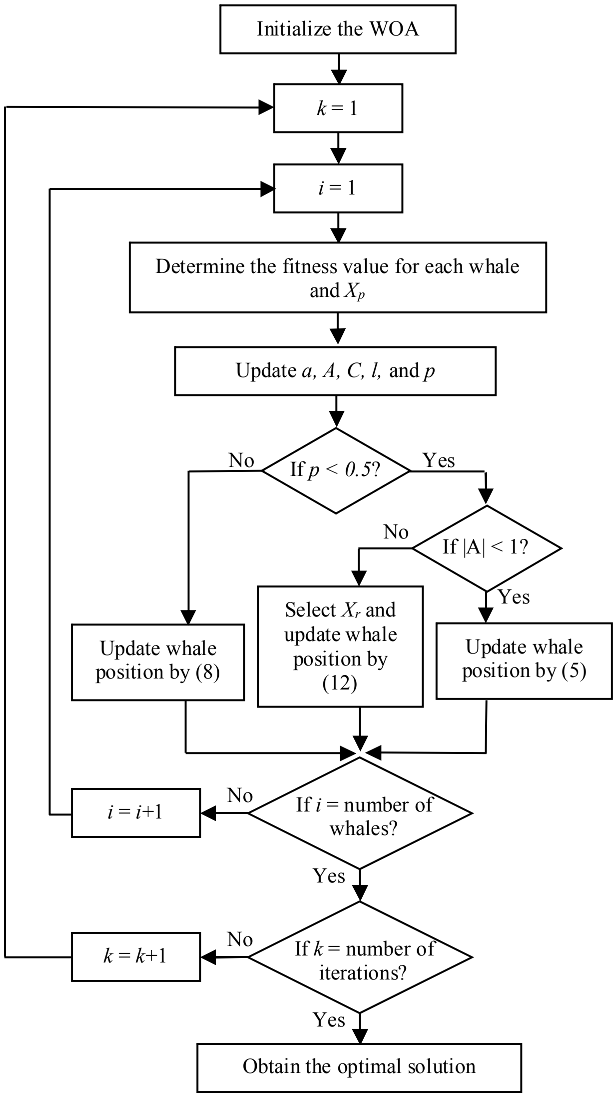

Fig. 4. Flowchart of the WOA.

Table 3
Design variables values and levels.

Design variables	k_{p1}	T_{i1}	k_{p2}	T_{i2}
Level (−1)	8	0.2	0.008	0.0015
Level (0)	10	0.6	0.010	0.0020
Level (1)	12	1	0.012	0.0025

composite design for obtaining accurate results. The MPOS and T_s of the V_{PCC} are chosen as an objective function and a nonlinear constraint, respectively. The WOA is applied to solve this optimization problem.

4.2. The WOA

The humpback whales dive deeply, create bubbles in a spiral shape around the prey, and swim to the surface during the maneuver. They usually attack the small fishes close to the surface. The mathematical model of WOA describes actions of encircling the prey, establishing spiral bubbles maneuver, and search for the prey. It can be modeled as follows:

Table 4
The simulation runs based on the RSM.

Run no.	k_{p1}	T_{i1}	k_{p2}	T_{i2}	MPOS (%)	T_s (s)
1	8	0.2	0.008	0.0015	8.85	1.2
2	12	0.2	0.008	0.0015	10.3	5.6
3	8	1	0.008	0.0015	3.9	0.42
4	12	1	0.008	0.0015	8	6.2
5	8	0.2	0.012	0.0015	9.5	1.3
6	12	0.2	0.012	0.0015	17.5	8.5
7	8	1	0.012	0.0015	5	0.43
8	12	1	0.012	0.0015	8.5	7.5
9	8	0.2	0.008	0.0025	10.5	2.9
10	12	0.2	0.008	0.0025	5.6	2.2
11	8	1	0.008	0.0025	6.6	2.4
12	12	1	0.008	0.0025	3	0.5
13	8	0.2	0.012	0.0025	10.5	2.8
14	12	0.2	0.012	0.0025	5.8	2.9
15	8	1	0.012	0.0025	6.8	2.2
16	12	1	0.012	0.0025	3	0.5
17	8	0.6	0.010	0.0020	3.7	0.82
18	12	0.6	0.010	0.0020	6.7	0.43
19	10	0.2	0.010	0.0020	7.75	1.5
20	10	1	0.010	0.0020	3.55	0.49
21	10	0.6	0.008	0.0020	3.6	0.44
22	10	0.6	0.012	0.0020	3.9	0.44
23	10	0.6	0.010	0.0015	4.1	0.51
24	10	0.6	0.010	0.0025	3.5	1.25
25	10	0.6	0.010	0.0020	3.6	0.44
26	10	0.6	0.010	0.0020	3.6	0.44
27	10	0.6	0.010	0.0020	3.6	0.44
28	10	0.6	0.010	0.0020	3.6	0.44
29	10	0.6	0.010	0.0020	3.6	0.44
30	10	0.6	0.010	0.0020	3.6	0.44
31	10	0.6	0.010	0.0020	3.6	0.44

4.2.1. Encircling the prey

In the WOA, the prey position is considered as a candidate solution. The encirclement of the humpback whales to the prey can be formulated by the following formulas:

$$D = |C.X_p(t) - X(t)| \quad (4)$$

$$X(t+1) = X_p(t) - A.D \quad (5)$$

where $X(t)$ is the whales position vector, t represents the current iteration, $X_p(t)$ is the prey position vector, and A and C are coefficient vectors, which can be calculated as follows:

$$A = 2a.r - a \quad (6)$$

$$C = 2.r \quad (7)$$

The vector a is linearly decreased from 2 to 0 with consistently of iterations and r is a random vector, which lies in the range of 0 and 1.

4.2.2. Bubble attacking of the prey

The attacking process points out exploitation or local search of the WOA. Two approaches are used to describe the bubbles behavior of the whales for attacking the prey. These are mathematically modeled as follows:

1. Shrinking encircling mechanism: in this approach, the whales swim around the prey in shrinking circles. This can be achieved by decreasing a from 2 to 0 with iterations and $|A| < 1$.
2. Spiral updating position: the humpback whales swim up to the prey in a spiral shape maneuver. The whales position can be updated by the following equations:

$$X(t+1) = D'.e^{bl}.\cos(2\pi l) + X_p(t) \quad (8)$$

$$D' = |X_p(t) - X(t)| \quad (9)$$

where b is a constant that determines the spiral logarithmic shape, and l is a random number lies in the range of −1 and 1.

The whales exhibit the two approaches simultaneously during the attacking process. Therefore, it is assumed that they do shrinking encircling mechanism by a probability of 50% and the spiral model by the same probability to update their positions, which can be modeled as follows [20]:

$$X(t+1) = \begin{cases} X_p(t) - A.D & \text{if } p < 0.5 \\ D'.e^{bl}.\cos(2\pi l) + X_p(t) & \text{if } p \geq 0.5 \end{cases} \tag{10}$$

where p is the probability number $\in$ [0,1].

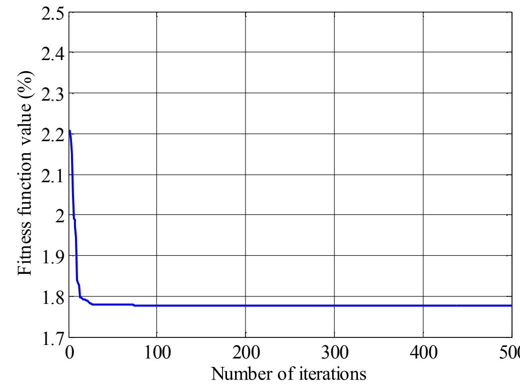

Fig. 5. Convergence of the fitness function value.

4.2.3. Searching for the prey

In this process, the humpback whales search for the prey and this presents the exploration or global search of the WOA. It is found that

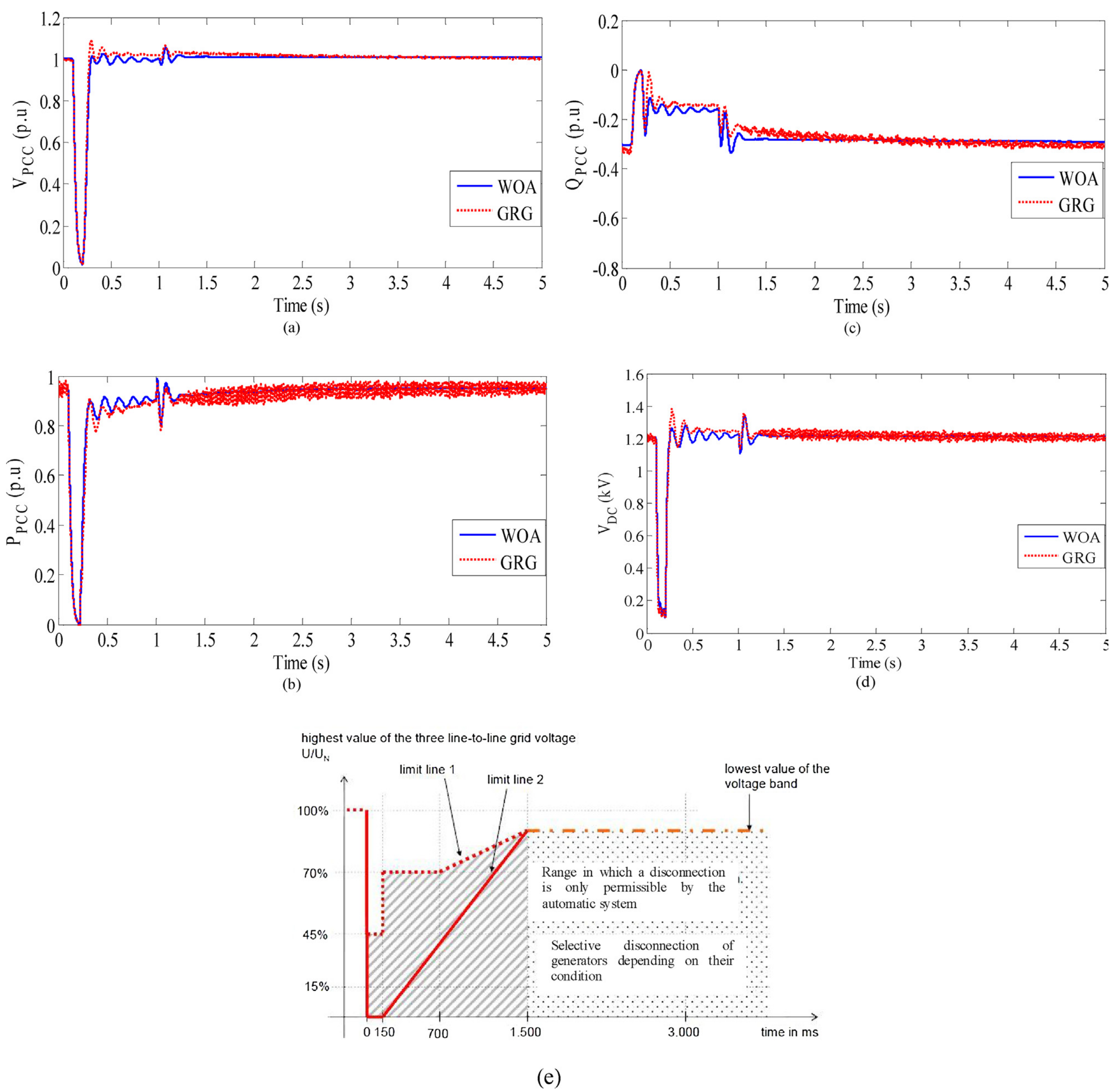

Fig. 6. PV system responses under a 3LG fault condition. (a) V_{PCC}. (b) P_{PCC}. (c) Q_{PCC}. (d) V_{DC}. (e) E. On Netz characteristics [30].

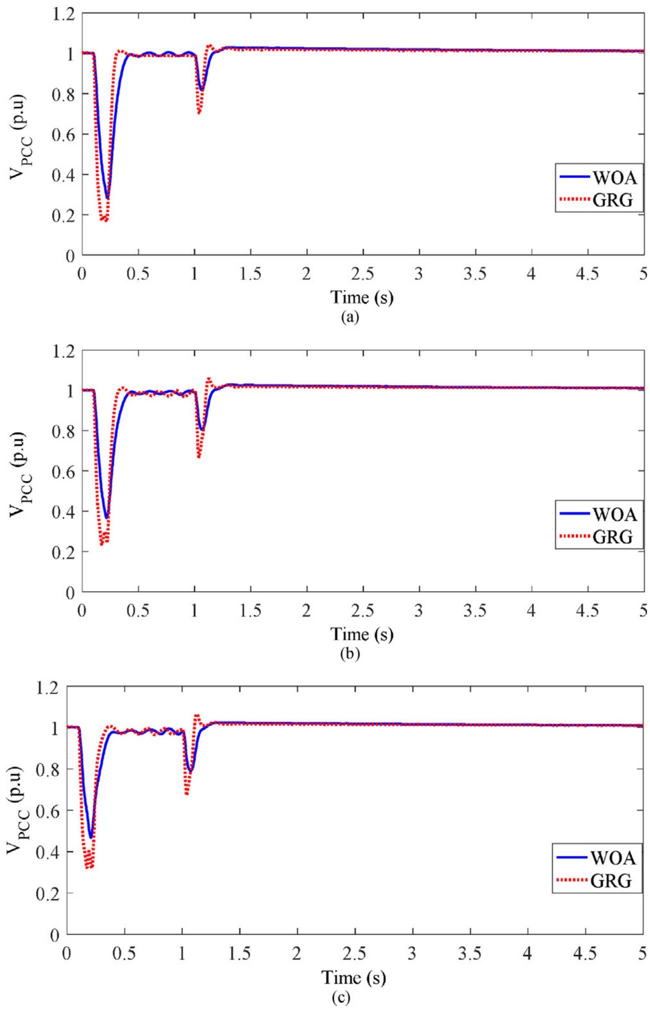

Fig. 7. Voltage response under different unsymmetrical fault conditions. (a) Double line to ground fault. (b) Line to line fault. (c) Single line to ground fault.

$|A| > 1$ can express the search process. The whales or the agents position can be updated as follows:

$$D = |C.X_r(t) - X(t)| \tag{11}$$

$$X(t+1) = X_r(t) - A.D \tag{12}$$

where $X_r(t)$ is a random whale position vector.

A flowchart of the WOA is indicated in Fig. 4. It starts with a random population of the humpback whales within the search space and terminates by X_p.

5. Design procedure

The proportional gain (k_p) and integral time constant (T_i) of the PI controller are selected as the design variables of the optimization problem, where their values affect the system responses. k_{p1}, T_{i1}, k_{p2}, and T_{i2} are the parameters of the PI-1 and PI-2 controllers, respectively. Each design variable has three levels, where the level −1 is its minimum value, the level 0 is its average value, and the level 1 is its maximum value. The values and levels of these design variables are listed in Table 3. Based on the standard central composite design of the RSM, 31 simulation runs are created, as illustrated in Table 4 and these runs are performed using PSCAD/EMTDC. The values of the MPOS and T_s of the V_{PCC} for each run are stored in Table 4. Then, the RSM model is built using the MATLAB environment and can be written by the following equations:

$$\begin{aligned} MPOS = {} & 3.5860 + 0.1694k_{p1} - 2.1083T_{i1} + 0.5639k_{p2} - 1.1306T_{i2} \\ & + 0.0219k_{p1}T_{i1} + 0.3719k_{p1}k_{p2} - 2.1281k_{p1}T_{i2} \\ & - 0.3906T_{i1}k_{p2} + 0.4844T_{i1}T_{i2} - 0.5656k_{p2}T_{i2} \\ & + 1.6303k_{p1}^2 + 2.0803T_{i1}^2 + 0.1803k_{p2}^2 + 0.2303T_{i2}^2 \end{aligned} \tag{13}$$

$$\begin{aligned} T_s = {} & 0.83 + 1.1033k_{p1} - 0.4589T_{i1} + 0.2617k_{p2} - 0.7783T_{i2} \\ & - 0.1094k_{p1}T_{i1} + 0.3181k_{p1}k_{p2} - 1.7906k_{p1}T_{i2} \\ & - 0.1556T_{i1}k_{p2} - 0.1969T_{i1}T_{i2} - 0.2444k_{p2}T_{i2} \\ & + 0.5599k_{p1}^2 + 0.9299T_{i1}^2 + 0.3749k_{p2}^2 + 0.8149T_{i2}^2 \end{aligned} \tag{14}$$

The WOA is applied to the RSM model and its code is built using MATLAB environment. The MPOS model represents the objective function and the T_s model is the nonlinear constraint function. The constraints of the problem include $8 \le k_{p1} \le 12$, $0.2 \le T_{i1} \le 1$, $0.008 \le k_{p2} \le 0.012$, $0.0015 \le T_{i2} \le 0.0025$, and $T_s \le 1$ s. The optimal characteristics of the WOA consists of 30 whales and 500 iterations. The optimization process is repeated more than 20 times to obtain the best fitness value and reveal the algorithm robustness. The best fitness value is 1.77% and the convergence of the fitness value is depicted in Fig. 5. The optimal level and size of the design variables are indicated in Table 5. Under these optimal conditions, the T_s is 0.498 s. The same procedure is repeated to design the PI-3 controller. The fitness function is the voltage error between V and the maximum voltage of the PV system. The design variables are k_{p3}, and T_{i3}. The RSM has 13 simulation runs, which are carried out using PSCAD/EMTDC. The WOA is applied to minimize the fitness function and achieve the optimal PI-3 controller parameters, as listed in Table 5.

Table 5
Optimal level and size of the design variables.

Design variables	k_{p1}	T_{i1}	k_{p2}	T_{i2}	k_{p3}	T_{i3}
Optimum level using WOA	0.6263	0.3639	−0.246	1	0.52	0.25
Optimum size using WOA	11.25	0.75	0.0095	0.0025	4.5	0.3
Optimum level using GRG	−0.99	−0.51	0.006	0.003	0.51	0.24
Optimum size using GRG	8	0.4	0.010	0.002	4.5	0.3

Table 6
Optimal characteristics of the GRG algorithm [16].

Number of iterations	500
Fitness function tolerance	1.10^{-6}
User-supplied derivatives	Hessian sparsity pattern
Minimum perturbation in derivatives	1.10^{-8}
Maximum perturbation in derivatives	0.1

6. Simulation results and discussion

The complete model of the grid-connected PV system including the switching model of its power converters is presented in details through this study for achieving realistic responses. The simulation results are carried out using PSCAD/EMTDC environment. In the simulation process, the time step and channel plot step are selected as 5 and 100 μs, respectively with the purpose of obtaining accurate results. Different scenarios are employed to check the effectiveness of the proposed WOA-based PI control strategy, as appear in the following subsections:

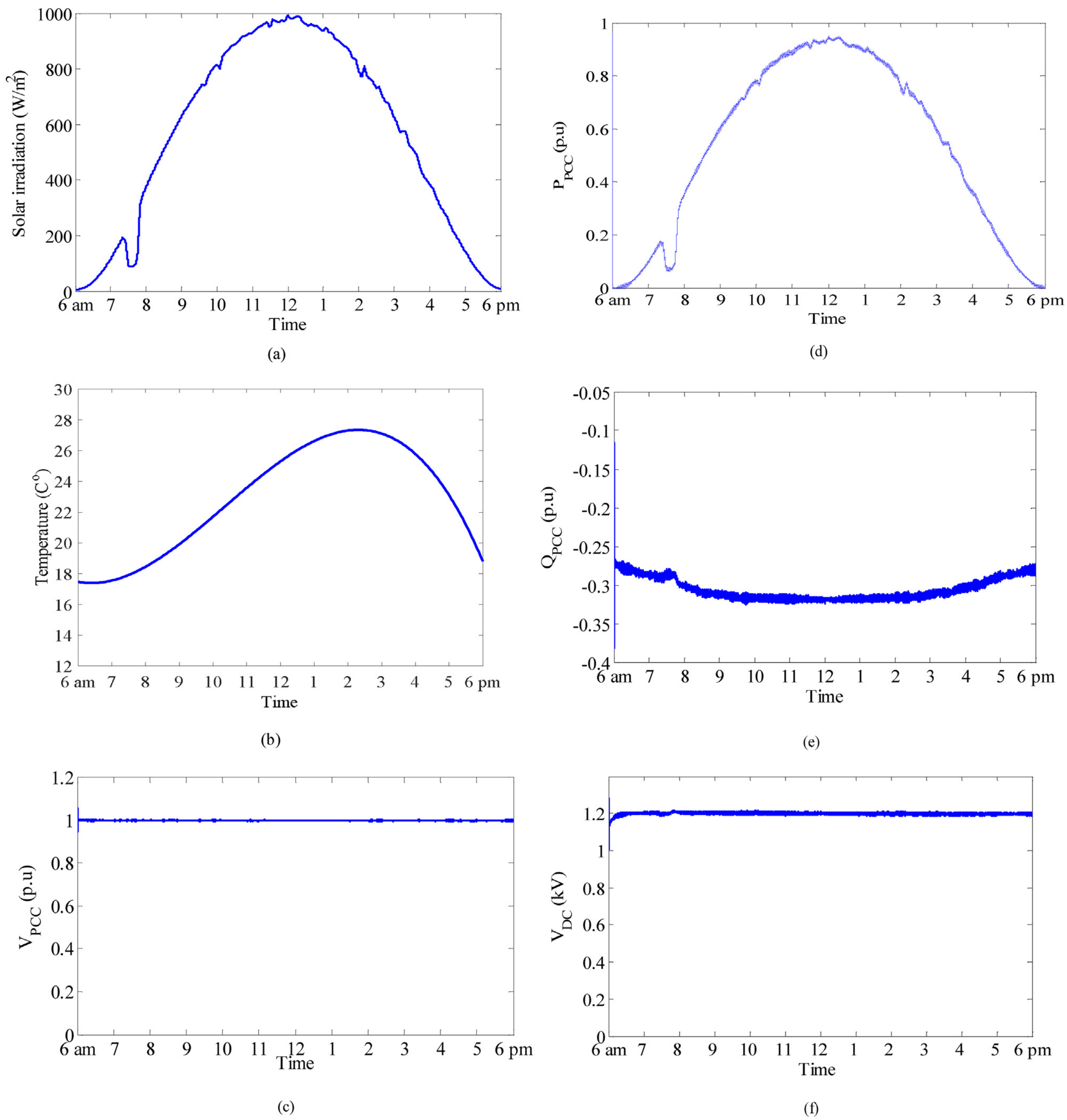

Fig. 8. PV system responses under different irradiation and temperature conditions. (a) Solar irradiation. (b) Temperature. (c) V_{PCC}. (d) P_{PCC}. (e) Q_{PCC}. (f) V_{DC}.

6.1. *Performance assessment of the PV system under a network disturbance*

This scenario aims at demonstrating performance assessment of the grid-connected PV system under a network disturbance using the proposed WOA-based PI control strategy. This disturbance is represented by a symmetrical three-line to-ground (3LG) fault, which happens at time t equals 0.1 s and its duration is 0.2 s at the point F, shown in Fig. 2b. The circuit breakers of the faulted line are opened to clear fault at t equals 0.2 s and reclosed again at t equals 1 s. Fig. 6a points out the response of the V_{PCC}. It can be noted that the V_{PCC} falls rigorously from its rated value because of the fault effect. The proposed control strategy can efficiently control the grid-side inverter to support the grid with enough reactive power. This causes a fast voltage recovery to the pre-fault value. For a fair comparison, the system responses using the WOA-based PI control strategy are compared with that obtained by using the conventional generalized reduced gradient (GRG)-based PI control scheme. The GRG algorithm is implemented using the MATLAB environment. The optimal characteristics of the GRG algorithm are listed in Table 6. The optimal level and size of the design variables using the GRG algorithm are indicated in Table 5. It is important to note here that the response of the V_{PCC} with the WOA-based PI controller is faster and better damped with minimum fluctuations than that obtained with the GRG-based PI controller. Furthermore, the proposed controller can efficiently control the DC chopper for a maximum power point tracking operation. The real power out of the PCC (P_{PCC}) response is illustrated in Fig. 6b. In addition, the responses of both the reactive power out of the PCC (Q_{PCC}) and V_{DC} are shown in Fig. 6c and d. It can be realized that all these responses using the proposed controller possess lower time transient specifications, lower fluctuations and steady state error. On the other hand, the V_{PCC} response using the proposed controller satisfies the low-voltage ride-through capability requirement by E. On Netz [30], as demonstrated in Fig. 6e. This ensures that the PV system remains in the grid-connected mode during the grid disturbance and supports the grid stability and reliability.

In addition, a fair comparison is made for the PV system responses using the proposed WOA-based PI controller and the GRG-based PI controller under different unsymmetrical faults such as double line to ground, line to line, and single line to ground faults. The V_{PCC} response under these conditions is demonstrated in Fig. 7a–c. It is worth for noting here that the V_{PCC} response with the WOA-based PI controller is faster and better damped with minimum fluctuations than that obtained with the GRG-based PI controller. This reflects the robustness of the proposed controller for dealing with symmetrical and unsymmeterical fault conditions.

6.2. Performance assessment of the PV system under different irradiation and temperature conditions

In this scenario, the performance of the grid-connected PV system is evaluated under different irradiation and temperature conditions using the designed WOA-based PI control strategy. The optimal PI controllers, whose parameters are listed in Table 5, are used in this study. For achieving a realistic study, the solar irradiation and temperature data that extracted from a field test are implemented. The real data are detected on 11 March 2015 at north of Riyadh city, Saudi Arabia. The simulation results are carried out using PSCAD/EMTDC for a complete sun day (6 am–6 pm). Fig. 8a points out the solar irradiation on the PV arrays' surface. The distribution of temperature of that day is demonstrated in Fig. 8b, where it varies from 17.5 to 27.5 °C. Fig. 8c indicates the response of the V_{PCC} with the WOA-based PI controller. It can be realized that this voltage response maintains constant at its rated value under these conditions. Moreover, the proposed controller successfully controls the DC chopper for a maximum power point tracking operation under these realistic conditions and the P_{PCC} response is depicted in Fig. 8d. The responses of Q_{PCC} and V_{DC} are shown in Fig. 8e and f, respectively. These responses are approximately fixed at their rated values under all operating conditions. This points out the high capability of the proposed control strategy to deal with the system nonlinearity and uncertainty under different operating conditions.

6.3. Performance assessment of the PV system under a sudden load disturbance in an autonomous mode

The main goal of this scenario is to point out the performance evaluation of the PV system under a sudden load disturbance in an autonomous mode of operation. The local load of the PV system is an inductive load and is initially operated at its rated value, then it is suddenly decreased to its half rated value at t equals 0.1 s. At t = 1 s, the load is returned back to its original value. As the load decreases, the load voltage transiently increases and the proposed controller is capable of controlling it around the rated value with minimum fluctuations, as shown in Fig. 9. As the load increases at t = 1 s, the load voltage transiently decreases and returns back to the rated value by the controller effect. This ensures the ability of the proposed controller to deal with variability and uncertainty of an islanded operation of PV systems.

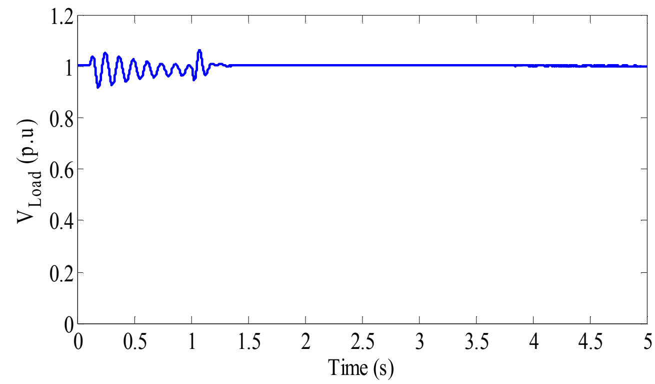

Fig. 9. Load voltage response under a sudden load disturbance in an autonomous mode of operation.

7. Conclusion

This paper has presented a novel application of the WOA-based PI controller with the purpose of improving the performance of PV systems. The fitted second order polynomial RSM model is developed to create the objective function and its constraints. The WOA is successfully applied to the RSM model to efficiently design the PI controllers of the PV system. The proposed controllers are used to control the converters of the PV system to achieve the targets. The simulation results have claimed that the PV system responses using the proposed controller are faster and better damped under different operating conditions such as (1) subject the system to symmetrical and unsymmetrical fault conditions, (2) study the system responses under different irradiation and temperature conditions using real irradiation and temperature data that extracted from a field test, and (3) subject the system to a sudden load disturbance in an autonomous mode of operation. Moreover, the effectiveness of the proposed controller is superior to that achieved using the GRG algorithm-based PI controller. Finally, it can be concluded that the dynamic performance of PV systems is enhanced with the WOA-based PI control strategy.

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